Project Planning

Unit_3

Project Scope Management

- **Definition:** The processes required to ensure that the project includes **all the work required, and only the work required**, to complete the project successfully.
- **Goal:** To define and control what is and is not included in the project. This prevents **scope creep** (uncontrolled changes or continuous growth in a project's scope) and ensures project success.

Key Processes:

•**Plan Scope Management:**
•Creates the **Scope Management Plan** – how the scope will be defined, validated, and controlled.
•Creates the **Requirements Management Plan** – how requirements will be analyzed, documented, and managed.
•**Collect Requirements:**
•Defining and documenting stakeholders' needs and expectations to meet project objectives.
•**Outputs:** Requirements Documentation, Requirements Traceability Matrix.
•**Define Scope:**
•Developing a detailed description of the project and product.
•**Output: Project Scope Statement** (describes project deliverables, boundaries, and acceptance criteria).

Key Processes:

•**Create WBS (Work Breakdown Structure):**
•**Decomposing** the project scope and major project deliverables into smaller, more manageable components.
•**Output: WBS** (hierarchical decomposition of total scope), **WBS Dictionary** (detailed description of each WBS element).
•**Validate Scope:**
•Formalizing acceptance of the completed project deliverables by the customer or sponsor.
•**Output:** Accepted Deliverables.
•**Control Scope:**
•Monitoring the status of the project and product scope.
•Managing changes to the scope baseline.
•**Output:** Change Requests, Work Performance Information.

Why is it Critical?

- **Clarity:** Ensures everyone understands what needs to be delivered.
- **Control:** Helps prevent unauthorized changes (scope creep).
- **Efficiency:** Focuses resources on necessary work, avoiding wasted effort.
- **Success:** Directly impacts project schedule, cost, and quality objectives.

Example: Community Health Post Construction

Project Goal:

To construct and commission a fully operational, earthquake-resistant, single-story Community Health Post (CHP) in the **Rural Municipality of Myagdi (Province 4)** within 12 months.

1. Define Scope (Project Scope Statement Excerpt):

Element	Description
Project Deliverables	A constructed health post building (2 consultation rooms, 1 pharmacy, 1 waiting area, 2 staff quarters, 2 toilets). Water supply connection (borehole or piped water, including a storage tank). Solar power system (5kW) for lighting and essential equipment. All necessary construction permits from the local *Wada Karyalaya* (Ward Office).
Project Boundaries (Inclusions)	Construction of the physical structure, utility hookups (water/power), and basic fencing. Procurement and installation of basic furniture (desks, chairs, examination couch).
Exclusions (What is NOT in scope)	Procurement of advanced medical equipment (X-ray, ultrasound). Staff recruitment and training (this is an operational scope, not the construction project scope). Construction of an access road (assumed existing path is sufficient).
Acceptance Criteria	Final inspection and approval by the Ministry of Health representative. Issuance of an occupancy certificate from the Rural Municipality Engineer.

2. Create WBS (Work Breakdown Structure) Snippet:

The WBS hierarchically breaks down the scope into manageable work packages.

Level	WBS Element	Type	Example Work Packages
1.0	**Community Health Post Project**	Project	
2.0	**Design & Approvals**	Deliverable	2.1 Topographical Survey (Hills), 2.2 Submit plans to *Wada*, 2.3 Acquire building permit.
3.0	**Site Preparation & Foundation**	Deliverable	3.1 Clear site (remove *jhadi*), 3.2 Excavation, 3.3 Pour plinth beam (earthquake-resistant design).
4.0	**Superstructure Construction**	Deliverable	4.1 Masonry wall construction (*Pathar/Itta*), 4.2 Roof truss assembly, 4.3 Roofing (tin or cement sheet).
5.0	**Utilities & Finishes**	Deliverable	5.1 Plumbing & sanitation, 5.2 Electrical wiring & solar panel install, 5.3 Interior plastering & painting (*Rang-Rogan*).
6.0	**Project Closeout**	Deliverable	6.1 Handover documents, 6.2 Final inspection.

3. Control Scope (Dealing with Scope Creep):

Imagine the following scenario:

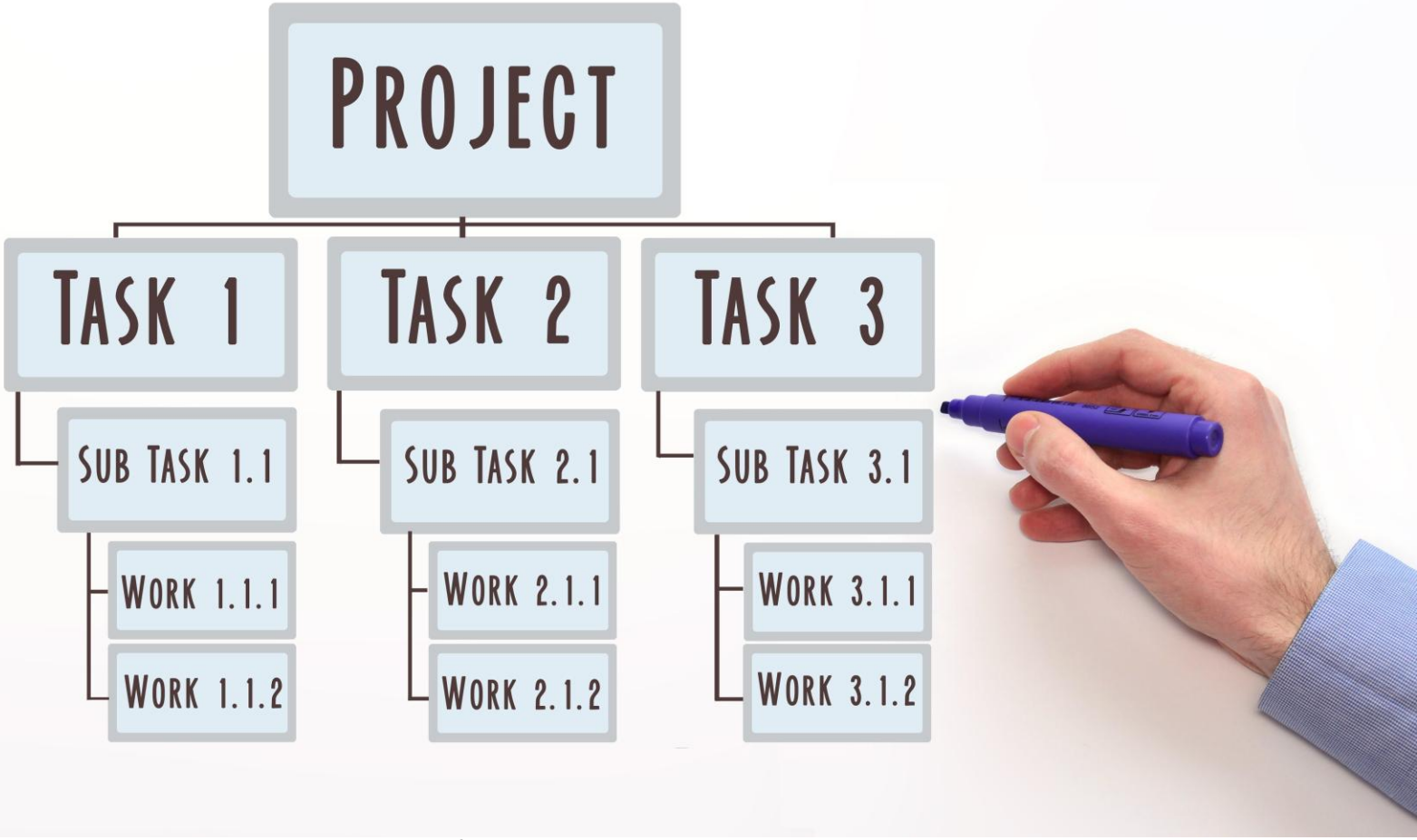
Issue: During construction, the local community elders request that a **separate kitchen/lunch room** be added to the staff quarters, citing long travel times for the nurses.

Project Manager's Response (Using Scope Control):

1. **Identify:** This is a **change request** as it falls outside the original Project Scope Statement and WBS (Exclusions section).
2. **Analyze:** The PM analyzes the impact:
   - **Cost:** Additional material and labor ($1,500 USD equivalent).
   - **Schedule:** 3-week delay to the overall schedule.
   - **Risk:** Requires a minor amendment to the approved building permit.
3. **Action:** The PM submits the request to the Project Sponsor (e.g., NGO head or Ministry official) via the **Change Control Process**.
4. **Decision:** The Sponsor approves the change because the long-term benefit of retaining staff outweighs the cost/schedule impact. The **Scope Baseline** is officially updated to include the new kitchen room.

If the change was denied, the PM would simply inform the stakeholders that the kitchen is **out of scope** for the current project and can be considered for a future phase.

WORK BREAKDOWN STRUCTURE



Work Breakdown Structure (WBS)

- Work Breakdown Structure (WBS) is a hierarchical structure to logically subdivide all the work elements of the project into a graphical presentation
- A project can be logically subdivided by project phases, location, tasks, sub-contractors etc.
- A **deliverable-oriented hierarchical decomposition** of the total scope of work to be executed by the project team.
- Breaks down the project into smaller, more manageable components.
- The lowest level of the WBS is called a **work package**.
- **Key Principle:** The WBS captures **100% of the project scope**, including all deliverables, internal and external.

Importance

- **Clarity & Understanding:** Provides a clear, visual representation of all project deliverables, ensuring everyone understands the full scope.
- **Scope Definition:** Helps define what is *in* and *out* of scope, preventing **scope creep**.
- **Improved Planning:** Facilitates detailed planning for schedule, cost, resources, and risks at a granular level.
- **Responsibility Assignment:** Easier to assign clear ownership and accountability for specific work packages.
- **Progress Tracking:** Enables more accurate monitoring and reporting of project progress against defined deliverables.
- **Estimation:** Allows for more precise cost and duration estimates for each work package.

Organizing a Cultural Event (e.g., Dashain Fair)

- Imagine a project to organize a **Dashain Cultural Fair** in a community ground in Kathmandu.

Understanding the WBS Levels:

- **Level 1 (Project):** "Dashain Cultural Fair" - The overall project.
- **Level 2 (Major Deliverables):** "Venue Setup," "Entertainment," "Food & Stalls," "Marketing," "Logistics" - Major components required for the fair.
- **Level 3 (Sub-Deliverables/Activities):**
  - Under "Venue Setup": "Stage Construction," "Seating Arrangement," "Decorations."
  - Under "Entertainment": "Cultural Dance," "Music Performances," "Kids Zone."
- **Level 4 (Work Packages):** The lowest level, representing specific, manageable tasks that can be assigned, estimated, and tracked.
  - Under "Stage Construction": "Order Materials," "Assemble Stage Frame," "Install Lighting."
  - Under "Cultural Dance": "Hire Dance Troupes," "Rehearsal Schedule."

Each work package has a clear deliverable. For example, "Assemble Stage Frame" delivers a complete stage frame.

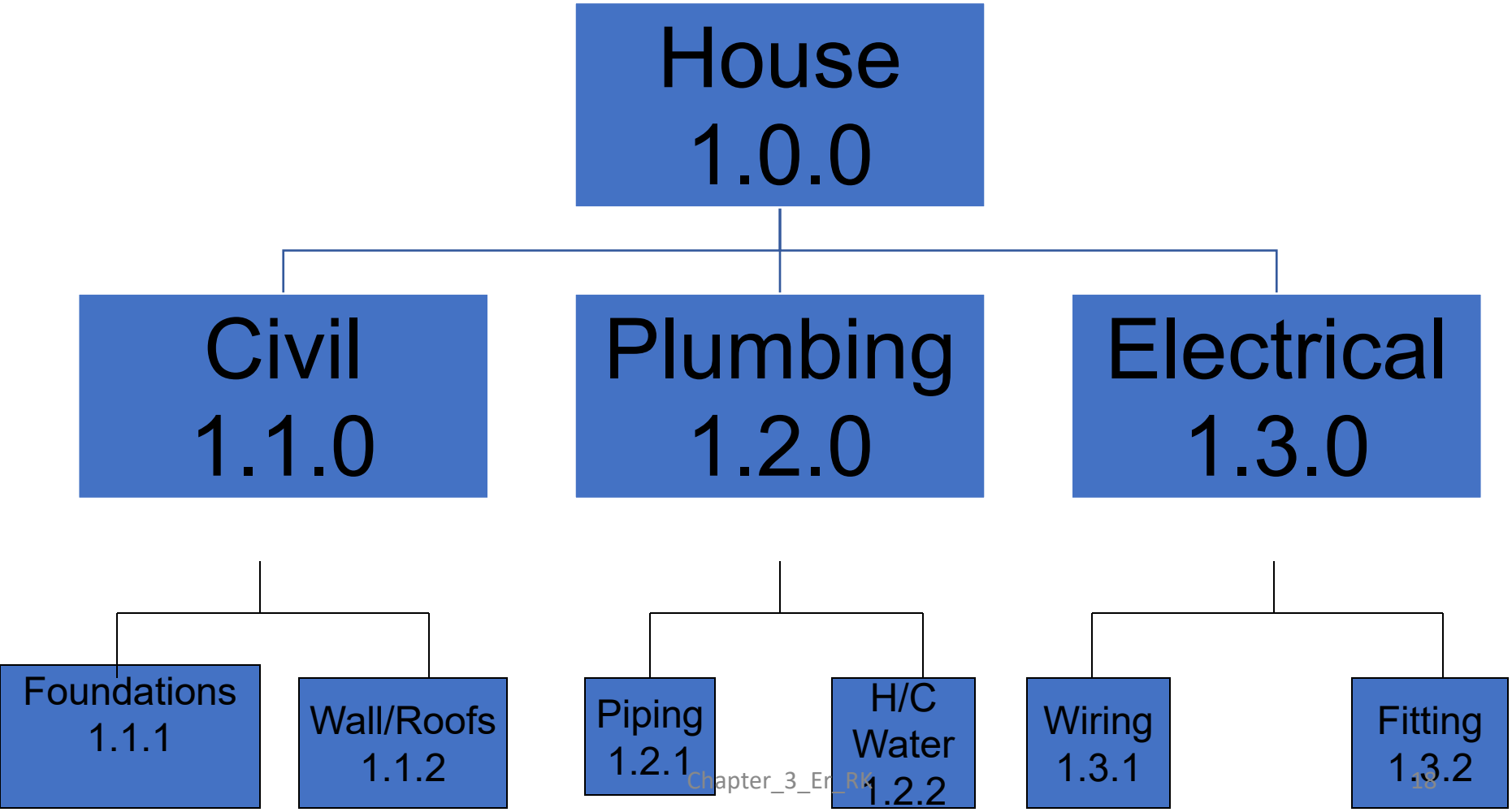
Components of Work Breakdown Structure

- Structure
- Description
- Coding
- Number of WBS Levels
- The level of details
- Roll-up

Uses of WBS

- It helps to visualize the scope of project clearly
- It helps to develop a complete lists of activities of the project
- It helps to estimate budget for the project

Work Breakdown Structure and Description



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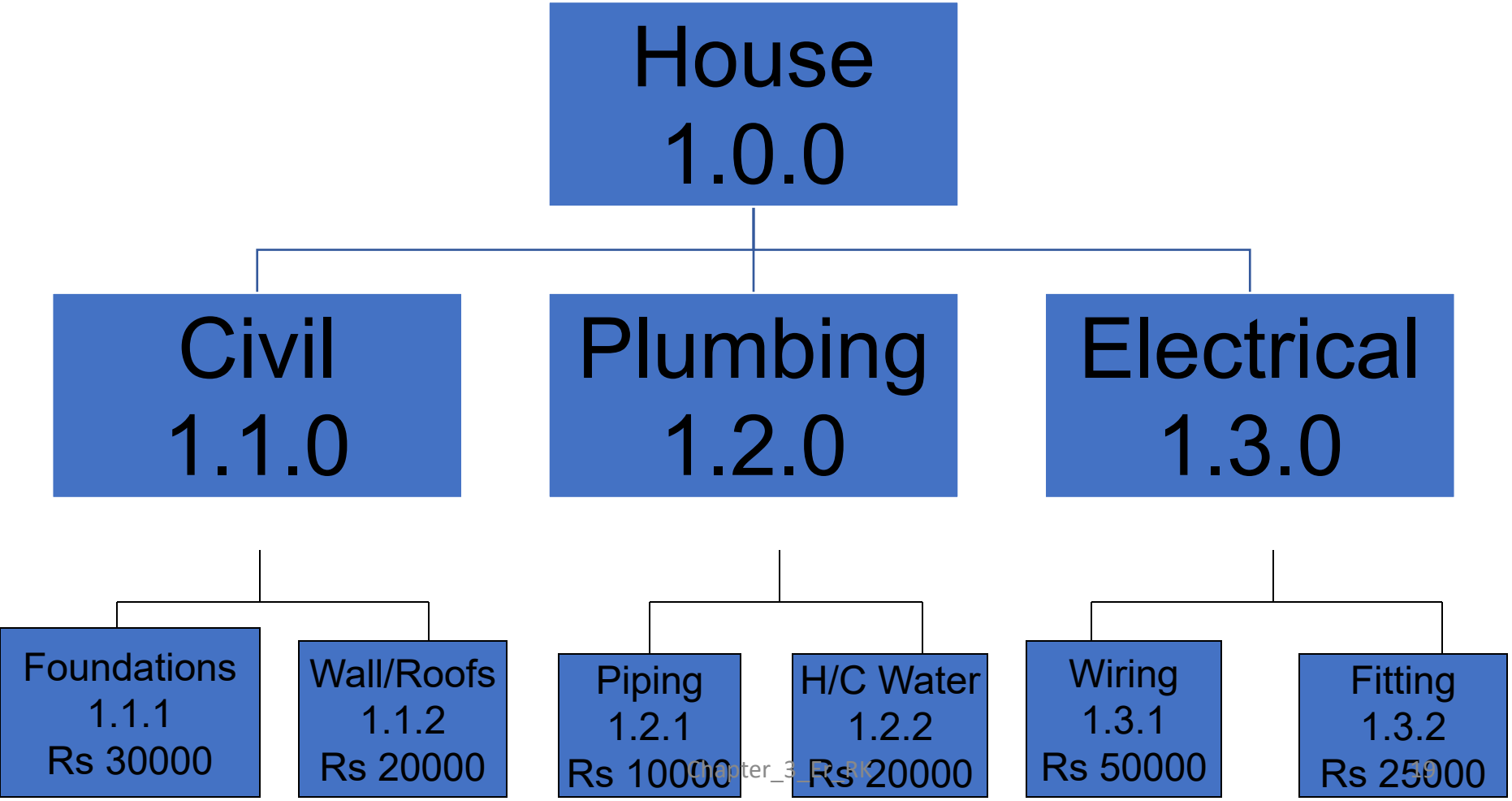
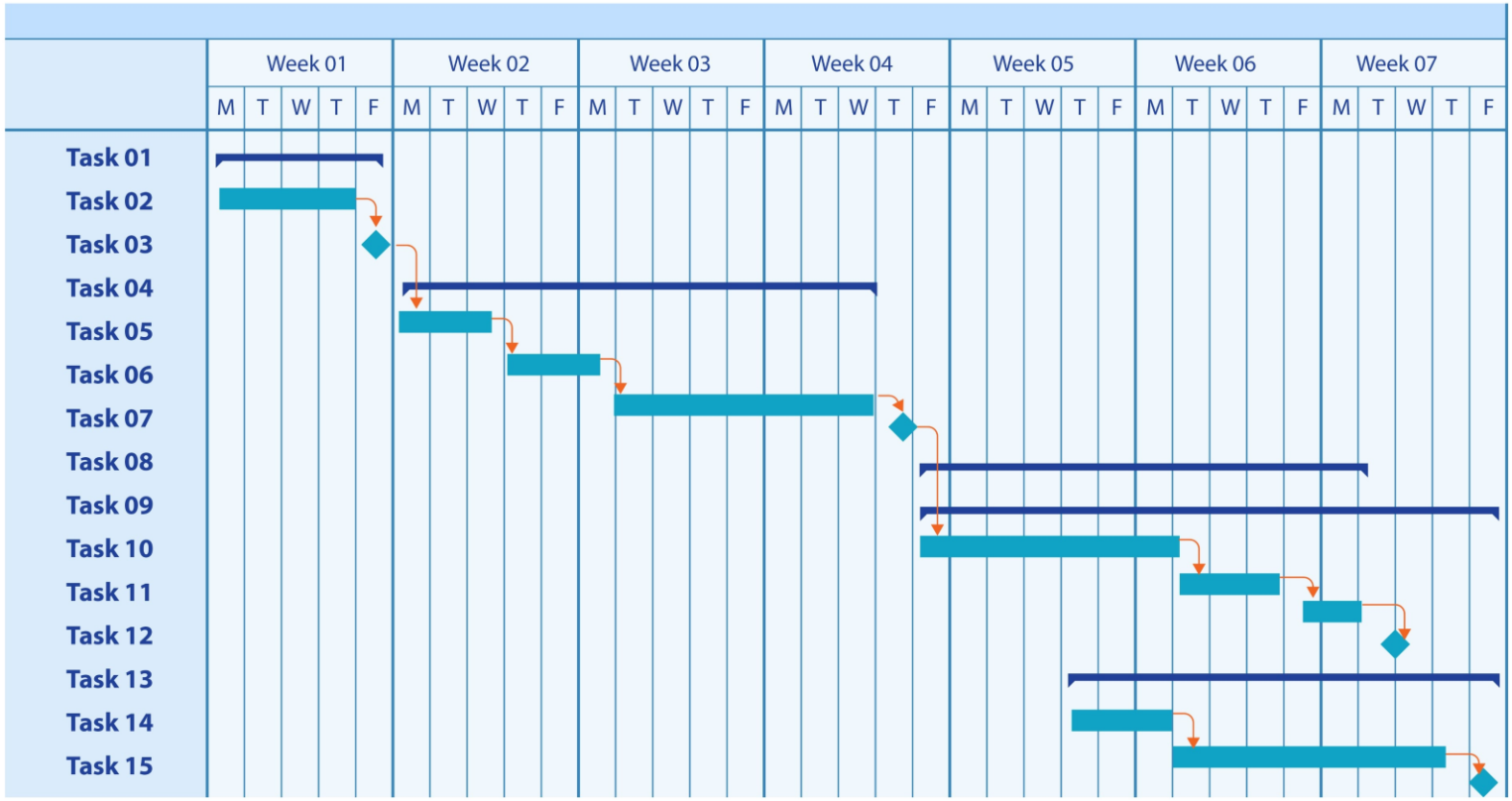


Time Management

- This involves the processes required to manage the **timely completion of the project**.
- **Scheduling:** Determining when project activities will be performed.
- **Gantt Charts:** A **bar chart** that shows the timeline for a project schedule, displaying the start and finish dates of the various elements.
- **Network Diagrams:** A graphical representation of the logical relationships (dependencies) among the project schedule activities. They show the sequence of work.
- **CPM (Critical Path Method):** A technique used to determine the **longest sequence of activities** in a project plan, which determines the shortest possible time to complete the project. The activities on this path are the **critical path** activities.
- **PERT (Program Evaluation and Review Technique):** A method for analyzing the tasks involved in completing a given project, especially useful when the time required for activities is uncertain. It uses optimistic, pessimistic, and most likely time estimates to calculate expected duration.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Task 13
Task 14
Task 15

Project Cost Management

- Project cost management ensures that the work is completed within the approved financial resources. It involves estimating all expenses, preparing a budget, and controlling costs throughout the project.
- It answers three essential questions:

1. How much will the project cost
2. When will the money be spent
3. How do we prevent overspending

Cost Management

- This includes the processes involved in **planning, estimating, budgeting, financing, funding, managing, and controlling costs** so that the project can be completed within the approved budget.
- **Budgeting:** Aggregating the estimated costs of individual activities or work packages to establish an authorized cost baseline.
- **Estimation Techniques:** Methods like **Analogous Estimating** (using past similar projects), **Parametric Estimating** (using mathematical models), and **Three-Point Estimating** (like PERT, using O, P, M estimates).

- Cost estimation is the process of predicting the financial resources required to complete all project activities. These include:
  - Labour
  - Materials
  - Equipment
  - Software and hardware
  - Travel
  - Training
  - Contingency reserves
  - Overheads and administrative costs
- A clear estimation supports planning, decision-making, and budgeting.

Common Estimation Techniques

A. Analogous Estimation

B. Parametric Estimation

C. Bottom-up Estimation

D. Three-Point Estimation (PERT Cost Method)

E. Expert Judgment

F. Vendor Bid / Tender Estimation

A. Analogous Estimation

- This method uses the cost of a similar past project as a reference.
- **Example**
  - A private IT firm in Kathmandu developed a school information system last year for NPR 8,00,000.
  - A new school in Pokhara now wants a similar system. Because the size and features are nearly identical, the team may estimate the new project to cost around the same amount, perhaps adding a small adjustment for inflation or added features.
- **Strength**
  - Quick and simple
- **Limit**
  - Less accurate because it depends on similarity.

B. Parametric Estimation

- Uses measurable units and applies a cost-per-unit factor.
- **Example**
  - A municipality plans to digitize 20,000 citizen records. If the cost to digitize one record is NPR 12 (from previous projects):
  - Calculation: 20,000 × NPR 12 = **NPR 2,40,000**
- This estimation works well when the work is repeatable and the per-unit cost is known.

C. Bottom-up Estimation

- Breaks the project into small work packages and estimates each item separately. All costs are then combined to find the total.
- Example: E-library System for a Rural Municipality

Work Package	Estimated Cost (NPR)
Requirement collection	40,000
UI/UX design	60,000
Backend development	1,50,000
Frontend development	1,20,000
Testing	50,000
Deployment	30,000
Training	20,000

Total project cost = NPR 4,70,000

This method is more accurate and is often used in formal contracts in Nepal.

D. Three-Point Estimation (PERT Cost Method)

Uses three values:

- **Optimistic (O)**
- **Most likely (M)**
- **Pessimistic (P)**

Expected cost is:

$$\text{Expected Cost} = \frac{O + 4M + P}{6}$$

Example

Cost of developing a new module for a hospital system:

- O = 80,000
- M = 1,00,000
- P = 1,60,000

$$\frac{80,000 + 4(1,00,000) + 1,60,000}{6} = \frac{80,000 + 4,00,000 + 1,60,000}{6} = \frac{6,40,000}{6} \approx 1,06,667$$

Expected cost ≈ **NPR 1,06,667**.

This helps when there is uncertainty or changing requirements.

E. Expert Judgment

- Cost is estimated using the experience of senior professionals who have handled similar projects.
- **Nepal Example**
- A government tender for a digital land-record system may rely on experts who understand software development, local infrastructure, and hardware procurement.

F. Vendor Bid / Tender Estimation

- Used when vendors submit financial proposals for a project.
  The project team compares bids and selects the most suitable offer based on **cost, quality, and capability**.
- **Example**
- The Nepal government's IT projects such as Nagarik App improvements or online payment gateways are often awarded through competitive bidding.

2. Budgeting

- Budgeting assigns estimated costs to project activities and produces the total approved project cost.
- It also determines when funds will be required at each stage.
- A proper budget includes:
  1. Activity cost estimates
  2. Contingency reserves (for known risks)
  3. Management reserves (for unknown risks)
  4. Cash-flow plan
  5. Cost baseline

Budgeting Example (Nepal Context)

A college in Pokhara plans to build a **Computer Lab**.

Category	Cost (NPR)
Computers (20 units × 45,000)	9,00,000
Furniture (tables, chairs)	1,50,000
Networking equipment	80,000
Electrical wiring	70,000
Software licenses	1,20,000
Labour and installation	60,000
Contingency reserve (10 percent)	1,08,000

Total Estimated Budget
NPR 15,88,000
The college prepares this as the cost baseline.
Monthly or weekly expenditure is then matched to the budget to monitor spending.

3. Cost Control

- Cost control ensures that the project does not exceed the budget.
  It includes monitoring spending, updating forecasts, approving variations, and reporting cost performance.
  - Common tools:
  - Earned Value Management
  - Cost variance analysis
  - Forecasting (Estimate at Completion)
  - Cost monitoring dashboards
- **Example**
- During the construction of an ICT lab, unexpected price increases in hardware may require a revised budget and approval from the college board.

Validating Examples (Simple Cases)

- **Example 1: Comparing Estimation Techniques**
- A school wants a small website.
- Analogous: Last year's website cost NPR 50,000 → estimate same.
- Parametric: 10 pages × NPR 4,000 per page = NPR 40,000
- Bottom-up:
  - Design: 10,000
  - Development: 22,000
  - Hosting: 5,000
  - Testing: 4,000
    Total = **NPR 41,000**
- Bottom-up is closest to the real value.

- **Example 2: Budget Revision Scenario**
- A municipality planned NPR 4,70,000 for an e-library.
  A new requirement added "SMSNotifications."
- Added cost: NPR 35,000.
- New budget
- 4,70,000+35,000=5,05,000
- Scope change increases the budget.

Risk Management

- Risk management is the process of identifying uncertain events that may affect a project, analyzing their impact, and taking actions to reduce their negative effects. It ensures that a project is prepared for unexpected situations.
  - A risk may cause:
  - Delay in schedule
  - Increase in cost
  - Reduction in quality
  - Failure to meet objectives
- Effective risk management increases the chance of project success.

1. Risk Identification

- Risk identification is the process of listing all possible risks that could harm the project. The goal is not to solve risks immediately, but to recognize them clearly.
- **Methods Used in Risk Identification**
  - Brainstorming
  - Expert interview
  - Checklists
  - Past project study
  - SWOT analysis (Strengths, Weaknesses, Opportunities, Threats)

Types of Risks to Identify

- **Technical Risks**
  - Failure of technology, performance issues, integration problems
- **Organizational Risks**
  - Staff shortage, poor coordination, change in leadership
- **External Risks**
  - Government policy changes, natural disasters, strikes
- **Project Management Risks**
  - Poor planning, unclear requirements, scope creep

Examples

- **Example 1: Hospital Management System Development (Nepal)**
- Possible risks identified:
- Delay in requirement approval from hospital administration
- Internet failure during system deployment
- Software integration issues with old hardware
- Sudden increase in patient data load beyond server capacity

- **Example 2: Construction of ICT Lab in a Public School**
- Risks:
- Suppliers may delay hardware delivery
- Electrical wiring may fail inspection
- Budget shortage
- Exams or festivals causing interruption in schedule

2. Risk Analysis

- Risk analysis evaluates the likelihood of each risk and the severity of its impact. The purpose is to prioritize which risks need immediate attention.
- There are **two main methods**:
  - A. Qualitative Risk Analysis
  - B. Quantitative Risk Analysis

A. Qualitative Risk Analysis

- Classifies risks as:
  - High
  - Medium
  - Low
- Based on probability and impact.

Example

Project: Online Municipal Tax System

Risk	Probability	Impact	Priority
Delay in government approval	High	High	High
Internet outage	Medium	Medium	Medium
Developer leaving job	Low	High	Medium
Data entry mistakes	Medium	Low	Low

High-priority risk requires immediate planning.

B. Quantitative Risk Analysis

- Uses numerical values to calculate the expected impact of risks.
  - Common tools:
  - Probability × Impact formula
  - Sensitivity analysis
  - Expected Monetary Value (EMV)

Example of Quantitative Analysis

A software server upgrade project identifies a risk:

•Probability of hardware failure: 20 percent

•Impact cost if failure occurs: NPR 3,00,000

Expected Monetary Value (EMV):

$$EMV = 0.20 \times 3{,}00{,}000 = 60{,}000$$

This means the project should reserve **NPR 60,000** to cover this risk.

3. Risk Mitigation Strategies

- Mitigation strategies reduce the impact or likelihood of risks.
- There are **four main response strategies**.
  - A. Risk Avoidance
  - B. Risk Reduction (Mitigation)
  - C. Risk Transfer
  - D. Risk Acceptance

- **A. Risk Avoidance**
- Eliminating the risk by changing the plan.
- **Example**
- If a web application feature depends on unstable third-party software, the project may decide to remove or replace that feature entirely.
- A municipality avoids using a small local SMS provider because of frequent outages. Instead, they use a more stable national provider.

- **B. Risk Reduction (Mitigation)**
- Taking actions to decrease the probability or impact.
- **Example**
- Adding additional testing to reduce the chance of software defects.
- To reduce the risk of power failure during deployment in rural schools, the team carries backup batteries and UPS systems.

- **C. Risk Transfer**
- Shifting the risk to another party, usually through contracts or insurance.
- **Example**
- A company outsources server hosting to a certified cloud vendor who guarantees uptime.
- A bank transfers cyber risk to a security firm responsible for 24-hour monitoring.

- **D. Risk Acceptance**
- The team acknowledges the risk and decides to deal with it if it occurs.
- Used when the cost of mitigation is higher than the risk impact.
- **Example**
- Minor graphical issues in the user interface are accepted because they do not affect system functionality.
- A small local startup accepts the risk of a slight delivery delay due to festival traffic.

Example of Full Risk Management Process

- **Project:** Deploying a Learning Management System (LMS) in a Pokhara College

- **Step 1: Identify Risks**
- Staff may resist using the new system
- Server may fail during peak load
- Internet connection may be unstable
- Extra training sessions may be required

Step 2: Analyze Risks

Risk	Probability	Impact	Priority
Server failure	Medium	High	High
Staff resistance	High	Medium	High
Internet issues	High	High	High
Extra training required	Medium	Low	Medium

Step 3: Mitigation Strategies

- Upgrade server capacity or use cloud hosting (Risk Reduction)
- Conduct early orientation for staff (Risk Reduction)
- Install backup internet connection (Risk Avoidance)
- Accept possible need for extra training (Risk Acceptance)

Communication Management

- Communication management ensures that the right information reaches the right people at the right time. It helps maintain coordination, reduce misunderstandings, and build trust among everyone involved in a project.
- A project fails more often because of poor communication than because of technical problems.

1. Stakeholder Engagement

- Stakeholder engagement is the process of identifying all individuals or groups that have an interest in the project and keeping them involved through proper communication.
- **Stakeholders may include:**
  - Project sponsor
  - Project manager
  - Team members
  - Government bodies
  - Clients and end users
  - Vendors and suppliers
  - Local communities (for public projects)
- The goal is to manage expectations, reduce conflict, and ensure everyone understands their responsibilities.

Steps in Stakeholder Engagement

- **A. Identify Stakeholders**
- Create a list of all people affected by the project.
- **Example :**
- Developing a digital registration system for a rural municipality.
  - Stakeholders include:
  - Municipal IT officer
  - Ward secretaries
  - Citizens using the system
  - Software developers
  - Data entry staff

Steps in Stakeholder Engagement

- **B. Analyze Stakeholder Interests**
- Not all stakeholders have the same influence or concern.
- Some may focus on cost, others on usability or security.
- Use a simple grid:

Stakeholder	Interest	Influence	Priority
Municipal Chief	System reliability	High	High
IT Officer	Technical performance	Medium	High
Citizens	Ease of use	High	Medium

Steps in Stakeholder Engagement

- **C. Plan How to Engage Stakeholders**
- Decide how often to communicate and what information to share.
- Examples:
  - Weekly meetings with the municipality
  - Monthly progress reports for the mayor
  - Daily internal updates for developers

Steps in Stakeholder Engagement

- **D. Manage and Monitor Engagement**
- As the project progresses, keep reviewing stakeholder needs.
- If expectations change, update plans accordingly.
- **Example:**
- Teachers in a Pokhara school request extra training when using a new LMS.
- The project team updates the schedule and adds a training session.

Example of Stakeholder Engagement

- **Case: ICT Lab Setup in a Public School**
  - Principal wants timely completion
  - Students want good-speed computers
  - Teachers want training sessions
  - Suppliers want clear payment schedules
  - Local government wants compliance with budget rules
- Engagement actions:
  - Weekly coordination meetings
  - Sharing installation plans with teachers
  - Providing a training workshop
  - Regular status updates to the municipality
- This prevents misunderstandings and keeps everyone aligned.

2. Information Distribution

- Information distribution is the process of sharing the right information with the right stakeholders using the right tools.
- The goal is to ensure transparency, smooth progress, and strong teamwork.

Methods of Information Distribution

- **A. Meetings**
- Daily stand-up meetings for the internal team
- Weekly review meetings with clients
- Monthly oversight meetings with government officials (if required)
- **Example:**
- In a software project for a hospital, the developer team meets every morning, while the hospital management receives a progress update every Friday.

Methods of Information Distribution

- **B. Reports**
- Reports summarize progress, risk status, budget updates, and upcoming tasks.
- Common project reports include:
  - Weekly status report
  - Monthly progress report
  - Risk report
  - Budget consumption report
- **Example:**
- A construction company working on a college building sends a monthly progress report to the college board with photos and financial updates.

Methods of Information Distribution

- **C. Communication Tools**
- These tools help share documents, updates, and messages.
- Examples:
  - Email
  - Zoom or Google Meet
  - Slack or Microsoft Teams
  - Project management tools (Trello, Jira, Asana)
  - WhatsApp for fast communication (widely used)
- **Example:**
- Developers working remotely from Kathmandu and Pokhara use Slack channels for code reviews and GitHub for file sharing.

Methods of Information Distribution

- **D. Documentation and Record Keeping**
- All meeting notes, design documents, contracts, and change requests must be recorded.
- This prevents disputes.
- **Example :**
- A municipality project stores all meeting minutes and agreements in a shared Google Drive folder and keeps physical copies for government audit.

Methods of Information Distribution

- **E. Dashboards and Visual Charts**
- Dashboards provide real-time project status through graphs and indicators.
- Useful for large projects with many stakeholders.
- **Example:**
- A hydropower project uses a dashboard showing construction progress and expenses so that the Ministry can monitor it remotely.

Example

- **Project: Developing an Online Entrance Registration System for a College**
- **Stakeholder Engagement Activities:**
  - Identify stakeholders: College management, IT staff, applicants, admission committee
  - Conduct initial meeting to collect expectations
  - Prepare a communication plan for updates
  - Schedule demonstration sessions for the committee
- **Information Distribution Activities:**
  - Weekly progress report to the college
  - Email notification to applicants during testing phase
  - Internal daily updates for the development team
  - Change requests recorded in Google Sheets
  - Training manual shared as PDF to staff
- This ensures the project moves smoothly without confusion.

Working Environment in Project Management

- The working environment describes **where project activities are performed** and **how teams collaborate**.
- The environment affects communication, productivity, cost, and project timelines.
- There are three common working arrangements:
  - **Onsite**
  - **Offsite**
  - **Work From Home (WFH)**
- Each environment has benefits and challenges.

1. Onsite Working Environment

- In an onsite setup, the project team works **physically at the client's or organization's location**.
- **Characteristics**
  - Direct access to clients, data, and equipment
  - Faster decision-making
  - Easier supervision and coordination
  - Suitable for projects requiring physical presence, such as installation, testing, or fieldwork

1. Onsite Working Environment

- **Advantages**
  - Immediate feedback from stakeholders
  - Strong teamwork
  - Better observation of real working conditions
  - Faster troubleshooting
- **Limitations**
  - Travel cost and travel time
  - Less flexibility
  - Limited working hours
  - Stress for team members if location is far or crowded

1. Onsite Working Environment

- **Example**
- A team installing a **Hospital Management System (HMS)** in a government hospital must work **onsite** to configure servers, train staff, and test real-time operations.

2. Offsite Working Environment

- Offsite work means the team works **from a different location**, usually from their own office or a dedicated project center, not at the client's site.
- **Characteristics**
  - Used when onsite access is not necessary
  - Communication happens through periodic meetings, emails, and online tools
  - Vendors can serve clients in different cities or countries

2. Offsite Working Environment

- **Advantages**
  - Lower travel costs
  - Quiet and controlled environment
  - Higher flexibility in planning team tasks
  - Access to internal company resources and support teams
- **Limitations**
  - Slower communication with the client
  - Difficult to understand client's day-to-day operations
  - Problems may take longer to resolve
- **Example**
  - A software company in Kathmandu develops an **Online Banking App for a Pokhara-based bank**. Most work (coding, design, testing) is done **offsite**, with meetings scheduled online.

3. Work From Home (WFH)

- In this arrangement, team members work from their homes using laptops and communication tools.
- WFH became popular globally due to COVID-19 and continues to be widely used in IT and research-based work.
- **Characteristics**
  - Employees work remotely
  - Work is coordinated through online tools and cloud systems
  - Suitable for coding, documentation, design, analysis, content creation

3. Work From Home (WFH)

- **Advantages**
  - Flexible working hours
  - Saves travel expenses and time
  - Helps maintain work-life balance
  - Useful for employees living far from office locations
- **Limitations**
  - Requires stable internet connection
  - Communication may be slower
  - Harder to monitor productivity
  - Lack of teamwork feeling due to isolation
- **Example**
  - During the COVID-19 lockdown, many IT firms in Lalitpur, Pokhara, and Bharatpur completed **mobile app development** and **digital services** through WFH.
  - Team members used Zoom, GitHub, Slack, and Google Drive to share work.

Comparative Table

Factor	Onsite	Offsite	Work From Home
Communication	Fast, direct	Good but scheduled	Mostly digital
Flexibility	Low	Medium	High
Cost	High (travel, setup)	Medium	Low
Monitoring	Easy	Moderate	Difficult
Suitable for	Testing, installation	Development, analysis	Coding, documentation

Example

- **Project: Developing and launching an E-learning System for a college in Pokhara**
- **Onsite Work**
  - Team visits the college to collect requirements
  - Tests the system with teachers and students
  - Conducts training sessions in the computer lab
- **Offsite Work**
  - Coding and UI design done at the company office in Kathmandu
  - Backend testing and database setup done in the development center
- **Work From Home**
  - Documentation writer prepares manuals from home
  - Part-time designer creates graphics remotely
  - Evening bug fixes done by remote developers
- This combination creates efficiency while managing cost and team comfort.

Gantt Charts

- A Gantt chart is a horizontal bar chart that represents a project schedule.
- It is a favored tool in project management for scheduling tasks over the duration of the project.